

Critical Analysis:

GoT: Effective Graph-of-Thought Reasoning in Language Models

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1 1. Methodological Strengths

1.1 A concrete representational hypothesis with an explicit mechanism

The study articulates a clear methodological hypothesis: linear Chain-of-Thought (CoT) rationales may under-represent non-sequential deductive structure in human reasoning, and an explicit graph representation of “thought units” can inject relational inductive bias into reasoning. The proposed Graph-of-Thought (GoT) mechanism operationalizes this claim by (i) constructing a discrete thought graph from the input and (ii) encoding this graph with a dedicated graph encoder before fusion with text (and vision) representations. This design moves beyond purely prompt-level changes and introduces a structured intermediate representation that can be tested via controlled ablations.

1.2 Reproducible construction protocol via Extract-Clustering-Coreference

A methodological strength is the explicit Extract-Clustering-Coreference (ECC) pipeline for graph construction. The pipeline extracts subject-verb-object triplets as raw nodes and edges, then applies coreference clustering and node replacement to densify the graph, followed by reconnection to form an adjacency matrix. The paper provides an explicit algorithmic description (Algorithm 1) and defines the graph as $G(N, E)$ with a binary adjacency matrix, which supports reproducibility and auditing of the graph extraction stage.

1.3 Modular architecture enabling attribution of gains

GoT uses separate encoders for text, vision (optional), and graph, then aligns features via cross-attention and integrates them using a gated fusion layer. This modularity is methodologically beneficial because it isolates the structural component (graph encoding) from the backbone language model (T5) and the vision encoder (ViT-large or DETR in additional experiments). Such separation helps attribute performance changes to the thought-graph pathway rather than to unreported backbone changes.

1.4 Systematic comparisons under matched backbones and controlled parameter checks

The experimental design includes strong and relevant baselines:

- For AQUA-RAT (text-only), comparisons include zero-shot and few-shot CoT prompting (GPT-3 text-davinci-002), Auto-CoT, Calcformer-T5-L, and a controlled two-stage CoT fine-tuning baseline using FLAN-Alpaca checkpoints.
- For ScienceQA (multimodal), comparisons include established VQA baselines, text-to-text LLM baselines (UnifiedQA, GPT-3.5), and Multimodal-CoT.

A notable methodological control is an enlarged Multimodal-CoT baseline whose parameter count is increased (via gated fusion MLP depth) to match GoT, showing that gains are not simply a parameter-count artifact.

1.5 Ablation studies targeting structure versus content

The ablation suite directly tests whether improvements derive from graph structure rather than the mere presence of extracted triplets:

- **Random Thought Graph:** preserves the GoT architecture but randomizes graph connectivity to disrupt structured deduction.
- **Triplets Concatenation:** appends extracted triplets to the input text, removing explicit structural encoding.
- **Coreference Injection:** replaces coreferent mentions in text without constructing a graph, probing whether coreference normalization alone explains gains.

On AQUA-RAT, GoT-T5_{base} achieves 32.09% accuracy, while Random Thought Graph yields 30.31% (a 1.78-point drop) and Triplets Concatenation yields 31.20% (a 0.89-point drop). These results provide direct evidence that structural encoding contributes beyond adding extracted information.

1.6 Transparency signals and partial reproducibility

The paper reports concrete training settings (100 epochs, learning rate 5×10^{-5} , weight decay 0.01, max input length 512, max number of nodes 150, batch size 10 for T5-base and 8 for T5-large, training on four NVIDIA A800 80G GPUs). It also reports inference throughput for Multimodal-CoT_{base} (16.33 eval samples per second) versus GoT (13.38 eval samples per second) and provides a public code repository link. These practices support partial reproducibility and enable independent verification of key claims.

2 2. Key Limitations

2.1 Limited benchmark coverage and constrained generalization claims

The empirical evaluation focuses on two datasets: AQUA-RAT (algebra word problems with rationales) and ScienceQA (multimodal multiple-choice science questions with lectures and explanations). While these benchmarks span text-only and multimodal settings, they do not establish broad generality to other reasoning tasks such as open-ended generation, long-horizon planning, tool-augmented reasoning, or domains with different discourse structure. The conclusions should therefore be interpreted as evidence of effectiveness under these benchmark distributions rather than a general reasoning improvement across tasks.

2.2 Reliance on external NLP tools without quantified extraction quality

ECC depends on Open Information Extraction and Stanford CoreNLP coreference resolution. The approach introduces a tool-dependent, discrete upstream stage, yet the paper does not quantify triplet extraction accuracy, coreference clustering quality, or error propagation rates. Without such measurements, it remains unclear whether GoT is robust under extraction noise, domain shift, or longer and more complex inputs where OpenIE and coreference resolution can degrade.

2.3 Rationale quality improvements are modest relative to answer accuracy gains

The paper reports that GoT has limited impact on rationale generation quality and attributes this to limited information in the text used for thought graph construction in the rationale stage. For AQUA-RAT, the reported ROUGE improvement for rationales is described as small (0.78 in ROUGE-L improvement in the first stage), whereas answer accuracy improves by 2.00 points for the base model. For ScienceQA, rationale generation ROUGE scores are high across methods, with GoT improving ROUGE to 94.39 compared to 93.24 for a rerun Multimodal-CoT_{base}, while answer accuracy increases from 85.19% to 87.59% for T5-base. This pattern suggests that GoT primarily improves final decision accuracy rather than substantially improving explanation fidelity.

2.4 Incomplete reporting of critical implementation details

Although core hyperparameters are provided, several reproducibility-critical details are not fully specified in the main text:

- Optimizer type, learning rate scheduling, gradient clipping, and decoding parameters for generation (such as beam size, temperature, top- k , top- p) are not clearly enumerated in the main experimental description.
- The exact OpenIE configuration, versioning, and any filtering heuristics for triplets are not exhaustively specified.
- The image captioning procedure is referenced for ScienceQA-style multimodal inputs, but caption generation settings are not fully detailed in the core method description.

These omissions make exact replication and fair third-party comparison more difficult.

2.5 Evaluation scope emphasizes accuracy, with limited robustness and failure-mode analysis

The evaluation is centered on multiple-choice accuracy (ACC) and rationale overlap metrics (BLEU, ROUGE, cosine similarity). There is limited systematic analysis of:

- Robustness to adversarial distractors, paraphrases, or extraction noise.
- Failure modes where graph construction introduces misleading edges or omits essential relations.
- Calibration and confidence, which are relevant when adding a structural reasoning module.

Appendices include case studies and attention visualization, but these analyses are not presented as systematic, statistically grounded evaluations.

2.6 Compute overhead is acknowledged but not fully characterized

The paper states that GoT increases computational costs and slightly slows training. Table-level reporting shows an increase in parameters from 227M to 233M (for the compared multimodal base setting) and a throughput drop from 16.33 to 13.38 eval samples per second. However, full cost characterization is incomplete: wall-clock training time, memory footprint, graph construction time per example, and scaling behavior with context length or node count are not comprehensively analyzed.

3. Technical Bottlenecks

3.1 Discrete graph construction as an information bottleneck

ECC converts raw text into a discrete graph via OpenIE triplets and coreference clustering, producing a binary adjacency matrix. This introduces two technical constraints:

- **Non-differentiability:** the structure is not learned end-to-end, preventing the model from correcting systematic extraction failures through gradient-based learning.
- **Information loss:** OpenIE triplets compress rich linguistic structure into subject-verb-object tuples, which may omit negation scope, quantifiers, temporal relations, and long-range dependencies that can be crucial in reasoning.

3.2 Graph attention locality and multi-hop reasoning trade-offs

The GAT encoder aggregates information over neighboring nodes. While stacking multiple layers can expand receptive fields, deeper message passing can introduce oversmoothing, increased compute, and instability, particularly when coreference merging densifies the graph. The method also uses a maximum node limit (150), which may force truncation or pruning on longer inputs, creating a scalability barrier for complex contexts.

3.3 Two-stage pipeline and error propagation

The framework generates rationales in Stage 1, then concatenates predicted rationales to the input for Stage 2 answer inference, reconstructing the thought graph accordingly. This design can propagate errors: incorrect rationales can become additional evidence for graph construction, and the second-stage graph may amplify spurious relations. Case-study discussion indicates that GoT can sometimes be robust to noisy rationales, but the pipeline remains vulnerable without systematic safeguards such as rationale verification or uncertainty-aware fusion.

3.4 Multimodal grounding bottlenecks via captioning and feature fusion

Multimodal GoT appends image captions to the input text for graph construction and aligns features via cross-attention, then fuses them through a gated mechanism. Caption-based injection can be lossy and may omit critical visual evidence. Additionally, the gated fusion design uses shared gating to combine modalities, which can create trade-offs between trusting text, image features, and graph features, particularly when one modality is unreliable. The ScienceQA configuration study with a weaker vision encoder (DETR) suggests that gains depend on how well the system can leverage language-derived structure when visual features are less robust, but this also highlights sensitivity to modality quality.

3.5 Efficiency constraints under deployment settings

Reported throughput indicates a measurable inference speed penalty. When combined with external tool calls for OpenIE and coreference resolution, real-world deployment could incur additional latency and engineering complexity. This can constrain applicability in interactive settings or large-scale batch inference.

4 4. Research Implications

4.1 Structured intermediate representations remain beneficial for smaller backbones

The reported improvements indicate that explicit structural inductive bias can improve accuracy even with relatively small backbones. On AQUA-RAT, GoT improves FLAN-Alpaca T5-base from 30.09% to 32.09% and T5-large from 33.73% to 34.48%. On ScienceQA, GoT improves Multimodal-CoT T5-base from 85.19% to 87.59% and T5-large from 90.45% to 90.81%. These results suggest that architectural structure can substitute for brute-force scaling in certain benchmark regimes.

4.2 Graph structure can improve multimodal reasoning beyond feature concatenation

ScienceQA results show consistent gains over Multimodal-CoT, a strong baseline that already integrates vision features into two-stage CoT reasoning. The improvements, including larger gains on image-related question classes reported in the analysis section, imply that graph structure may help align textual and visual evidence by providing relational scaffolding for cross-modal integration.

4.3 Benchmark success does not fully resolve the reasoning-versus-pattern-matching debate

GoT is motivated by human non-linear reasoning, yet the evaluation primarily measures answer accuracy and rationale overlap. Without targeted tests for deductive validity, counterfactual consistency, or graph faithfulness, it remains ambiguous whether improvements reflect genuine multi-hop deduction or more effective feature routing and denoising. This implication generalizes to the broader field: intermediate-structure methods require complementary evaluation protocols that directly assess reasoning behaviors.

4.4 Tool-mediated reasoning systems face a recurring challenge of upstream reliability

The dependence on OpenIE and coreference resolution mirrors a broader trend in hybrid NLP systems: pipeline components can introduce hidden brittleness. The GoT results indicate that structured pipelines can work well in-distribution, but robust deployment requires either more reliable upstream components or end-to-end learning that can compensate for upstream noise.

5 5. Potential Research Directions

5.1 Learned or differentiable graph construction

A concrete direction is to replace or augment ECC with learned relation extraction and coreference modules trained jointly with the downstream objective. Two actionable variants follow:

- **Soft adjacency:** predict edge probabilities rather than binary adjacency, enabling uncertainty-aware message passing and reducing brittleness from hard extraction decisions.
- **Joint training:** train a graph-construction head that optimizes downstream accuracy and rationale quality, while retaining ECC as an interpretable prior or initialization.

5.2 Graph quality and faithfulness evaluation protocols

Introduce intrinsic evaluation of the thought graph, including:

- Triplet correctness and coverage against human-annotated relational facts on a sampled subset.

- Coreference cluster consistency measures and error rates.
- Faithfulness checks linking decisive answer evidence to specific graph paths or node subsets.

Such protocols would strengthen claims about deductive reasoning and provide diagnostics when performance degrades.

5.3 Robustness stress tests and controlled perturbations

Develop stress tests that preserve semantics but challenge extraction and structure:

- Paraphrase and entity-renaming perturbations to test invariance of the constructed graph and final answers.
- Adversarial distractors that induce spurious triplets or misleading coreference links.
- Noise-injection experiments that systematically corrupt a fraction of extracted edges to quantify sensitivity.

5.4 Mitigating two-stage error propagation

Design mechanisms to reduce reliance on potentially incorrect Stage 1 rationales:

- Rationale verification modules that score or filter rationales before Stage 2 graph reconstruction.
- Multi-sample rationale generation with answer aggregation, combined with reuse or ensembling of graphs.
- Training objectives that enforce consistency between rationale content and answer choice.

5.5 Efficiency-oriented graph encoding and deployment considerations

Pursue computational optimizations aligned with the reported throughput gap:

- Node pruning or salience-based graph sparsification to reduce GAT compute.
- Caching graph construction outputs when inputs share common structure, such as repeated contexts.
- Distillation of the graph encoder into lightweight adapters or low-rank modules.

Future work should standardize reporting of wall-clock time, memory, and latency contributions from both neural and tool-based components.

5.6 Extending beyond deduction-only graphs

ECC focuses on a deduction template grounded in triplets. Future research can extend the graph schema to include richer relation types (temporal, causal, negation, quantification) and investigate whether these extensions improve higher-grade and commonsense-heavy ScienceQA subsets, where the paper reports smaller or unstable gains.

6 6. Conclusion

GoT contributes a well-motivated and modular method for injecting non-linear structure into CoT-style reasoning by constructing a thought graph with ECC, encoding it with a GAT-based GoT encoder, and integrating it through cross-attention and gated fusion in a two-stage rationale-then-answer framework.

Methodological strengths include explicit and reproducible graph construction, systematic comparisons under matched backbones, parameter-matched baseline controls, and ablations that isolate the role of structure versus content. Empirical results demonstrate consistent gains on AQUA-RAT (30.09% to 32.09% for T5-base) and ScienceQA (85.19% to 87.59% for T5-base), while also revealing practical constraints: dependence on external extraction tools without quantified error analysis, modest rationale-generation improvements relative to answer accuracy gains, limited benchmark breadth, and measurable efficiency overhead (throughput drop from 16.33 to 13.38 eval samples per second in the compared multimodal base setting). The most promising future directions include learned and uncertainty-aware graph construction, intrinsic graph-quality and faithfulness evaluation, robustness stress testing, mitigation of two-stage error propagation, and efficiency-driven graph encoding strategies with comprehensive cost reporting.